

Shift-invariant functions that behave like polynomials

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Abstract

In recent works [KK24, KK25] it was shown that the function $\chi : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ behaves like the polynomial $1 + X$ in the factor ring $\mathcal{R}_n = \mathbb{F}_2[X]/(m(X))$ where $m(X) = X^{(n+1)/2}$ if n is odd and $m(X) = X^n + X^{n/2}$ if n is even. More generally, the set $G_n = \gamma_0 + \text{span}\{\gamma_2, \gamma_4, \dots\}$ (for some functions γ_{2k} and $\chi = \gamma_0 + \gamma_2 \in G_n$) is an abelian monoid with respect to composition and is isomorphic to $1 + \text{span}\{X, X^2, \dots\} \subseteq \mathcal{R}_n$ under multiplication, which allows to deduce various properties about the maps in G_n , e.g. for which n they are invertible, what their iterates are and what their inverse is. We present these results and then generalize them to a much broader class of functions based on something we call quasi-conserved landscapes. We show that this class of functions is exactly the class of shift-invariant functions which behave like polynomials in the same way as χ .

Key Words : Shift-invariant permutations, Cryptography, Chi-function

1 Introduction

An important function in cryptography is the function $\chi : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ defined by

$$\chi(x)_i = x_i + (1 + x_{i+1})x_{i+2},$$

where y_i is the i -th coordinate of a vector $y \in \mathbb{F}_2^n$ for $i \in \{0, \dots, n-1\}$ and the indices are taken modulo n . The function χ is used in some important primitives, most notably SHA-3 [NIS15] and ASCON [DEMS21]. Due to its relevance there have been a number of papers that studied mathematical

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properties of χ , see for example [GKKK25, KK24, KK25, LDS⁺26, LSMI22, SD24a, SD24b].

In Section 2 we present the results of [KK24, KK25]. We keep the discussion brief and for more details refer to the cited papers. In Section 3 we present a generalization of those results to a much broader class of functions.

2 On the functions χ and γ_{2k}

We start by establishing some notation. We set $S : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ as the cyclic left-shift operator which is given by

$$S(x_0, x_1, \dots, x_{n-1}) = (x_1, \dots, x_{n-1}, x_0).$$

A function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is called shift-invariant if $S(f(x)) = f(S(x))$ for all $x \in \mathbb{F}_2^n$. Next, we let $\mathbb{1}$ be the vector $(1, \dots, 1) \in \mathbb{F}_2^n$. For vectors $x, y \in \mathbb{F}_2^n$ we define their component-wise product $z = x \odot y$ by $z_i = x_i y_i$ for $i \in \{0, \dots, n-1\}$. Note that the operation $\odot$ on $\mathbb{F}_2^n$ is associative, commutative and distributive over addition and has an identity $\mathbb{1}$. Observe that S is linear and $S(x \odot y) = S(x) \odot S(y)$ for all $x, y \in \mathbb{F}_2^n$. We extend $\odot$ to functions $f, g : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ by $(f \odot g)(x) = f(x) \odot g(x)$ for all $x \in \mathbb{F}_2^n$. With this notation we can write $\chi = \text{id} + (\mathbb{1} + S) \odot S^2$. The functions $\gamma_{2k} : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ for $k \geq 1$ are defined via

$$\gamma_{2k} = (\mathbb{1} + S) \odot (\mathbb{1} + S^3) \odot (\mathbb{1} + S^5) \odot \dots \odot (\mathbb{1} + S^{2k-1}) \odot S^{2k}.$$

If we set $\gamma_0 = \text{id}$ as the identity mapping, then $\chi = \gamma_0 + \gamma_2$. We set $\Gamma_n = \text{span}\{\gamma_0, \gamma_2, \gamma_4, \dots\}$ as the linear span of the functions γ_{2k} , it is a subspace of the space of all shift-invariant functions on $\mathbb{F}_2^n$. In [KK24] the following key lemma was shown.

Lemma 2.1. *Let $m \geq 2$ be even and $f = \gamma_0 + \sum_{i=1}^s \alpha_i \gamma_{2i}$ with $\alpha_i \in \mathbb{F}_2$ be arbitrary. Set $\alpha_0 = 1$. Then*

$$\gamma_m \circ f = \sum_{i=0}^s \alpha_i \gamma_{2i+m}.$$

As an example of Lemma 2.1 we see that

$$\gamma_4 \circ (\gamma_0 + \gamma_2 + \gamma_8) = \gamma_4 + \gamma_6 + \gamma_{12}$$

which is very similar to the polynomial multiplication

$$X^4 \cdot (1 + X^2 + X^8) = X^4 + X^6 + X^{12}.$$

We set

$$G_n = \gamma_0 + \text{span}\{\gamma_1, \gamma_2, \dots\}$$

and

$$\mathcal{R}_n = \mathbb{F}_2[X]/(m(X))$$

where

$$m(X) = \begin{cases} X^{(n+1)/2} & n \text{ is odd,} \\ X^n + X^{n/2} & n \text{ is even.} \end{cases}$$

We have a map $\varphi : \Gamma_n \rightarrow \mathcal{R}_n$ given by

$$\varphi \left(\sum_{i=0}^s \alpha_i \gamma_{2i} \right) = \sum_{i=0}^s \alpha_i X^i.$$

Furthermore we set

$$\mathcal{M}_n = \left\{ 1 + \sum_{i=1}^s \alpha_i X^i : \alpha_i \in \mathbb{F}_2 \right\} \subseteq \mathcal{R}_n.$$

It is easy to see that $\mathcal{M}_n$ is a monoid with respect to the multiplication in $\mathcal{R}_n$. For a monoid $(M, \star)$ with neutral element 1 we set

$$M^* = \{m \in M : \exists n \in M : m \star n = n \star m = 1\}$$

as the unit group of M . The following theorem is then one of the main results of [KK24, KK25].

- Theorem 2.2.**
1. The set G_n is an abelian monoid under composition.
 2. $G_n \cong M_n$ as monoids and $G_n^* \cong M_n^*$ as groups, each under the isomorphism φ .
 3. If n is odd, then $G_n = G_n^*$, i.e. G_n is a group.
 4. If n is even, then $G_n^* \subsetneq G_n$.

For n odd Theorem 2.2 says that every function in G_n is a permutation of $\mathbb{F}_2^n$. For n even this is not the case, however the isomorphism to a polynomial factor ring gives an easy criterion which enables to check whether a function $f \in G_n$ is a permutation.

Theorem 2.3. Let n be even and $f = \gamma_0 + \sum_{i=1}^s \alpha_i \gamma_{2i} \in G_n$. Then f is a permutation on $\mathbb{F}_2^n$ if and only if

$$\gcd(1 + \sum_{i=1}^s \alpha_i X^i, X^n + X^{n/2}) = 1.$$

If f is a permutation, then let $g(X) = \sum_{i=0}^t \beta_i X^i$ be the inverse of $1 + \sum_{i=1}^s \alpha_i X^i \in \mathbb{F}_2[X]/(X^n + X^{n/2})$. Then $\sum_{i=0}^t \beta_i \gamma_{2i} \in G_n$ is the inverse of f .

Note that the inverse of $1 + \sum_{i=1}^s \alpha_i X^i \in \mathbb{F}_2[X]/(X^n + X^{n/2})$ can be easily computed using the Extended Euclidean Algorithm.

As powers of polynomials are well-understood and easy to compute (especially for binomials) we obtain the following result for the iterates of χ (and in general a similar result holds for other maps in G_n):

Theorem 2.4. *It holds that*

$$\chi^k = \sum_{i=0}^k \binom{k}{i} \gamma_{2i}.$$

Note that Lucas' theorem allows to compute $\binom{k}{i} \in \mathbb{F}_2$ easily.

As a corollary we obtain a formula for the inverse of χ which was previously found in [LSMI22] and the order of χ which was previously found in [SD24b].

Corollary 2.5. *Let n be odd. Then*

$$\chi^{-1} = \gamma_0 + \gamma_2 + \dots + \gamma_{n-1}.$$

The order of χ is 2^m where $2^m < n < 2^{m+1}$, i.e. $\text{ord}(\chi) = 2^{\lceil \log_2 n \rceil}$.

Combined with Theorem 2.4 this gives knowledge on the cycle structure of χ .

Theorem 2.6. *Let n be odd. All cycles of χ have length $k = 2^s$ for some $0 \leq s \leq m$. The number of nonzero elements of order $\leq 2^s$ is given by $c(2^s, n)$ with*

$$c(k, n) = |\{x \in \mathbb{F}_2^n \setminus \{0\} : \gamma_{2k}(x) = 0\}|.$$

Using tools from enumerative combinatorics, in particular combinatorics on words, one can determine these numbers $c(k, n)$ by finding a generating function. Specifically, by applying [Sta11, Proposition 4.7.13] to a particular monoid we obtain:

Theorem 2.7. *For fixed k we have the generating function*

$$\sum_{n \geq 0} c(k, n) x^n = \frac{x B'_k(x)}{1 - B_k(x)}$$

with the polynomial $B_k(x) = x + 2x^3 + 4x^5 + \dots + 2^{k-1}x^{2k-1}$. Furthermore,

$$c(k, n) = \sum_{N=0}^{n \cdot \frac{k-1}{2k-1}} \frac{n 2^N}{n - 2N} \sum_{q=0}^{N/k} (-1)^q \binom{n - 2N}{q} \binom{n - N - qk - 1}{n - 2N - 1}.$$

Theorem 2.7 does not appear in [KK24, KK25]. One consequence of Theorem 2.7 is that we can compute that for large k it holds that $\gamma_{2k}(x) = 0$ for most $x \in \mathbb{F}_2^n$, which allows to approximate χ^{-1} by a function with smaller degree. An earlier version of [LNV⁺25] indirectly used such an approximation to attack CHiLow [BDG⁺25] and it is possible that this idea also allows other attack on primitives based on χ , such as Rasta [DEG⁺18]. However, this requires more research.

Moving away from χ , we can look at the other maps in G_n . Generally, when we fix a shift-invariant function f then an interesting question is for which n it is a permutation on $\mathbb{F}_2^n$. In other words, we want to determine the set

$$\text{inv}(f) = \{n \in \mathbb{N} : f \text{ is a permutation on } \mathbb{F}_2^n\}$$

which has been introduced in [OS22]. Proposition 6.1 combined with the subsequent discussion in [Dae95] shows that there is a set $\xi(f)$ such that $n \in \mathbb{N} \setminus \text{inv}(f)$ if and only if n is not a multiple of an element in $\xi(f)$. With this notation, the statement that χ is invertible if and only if n is odd is equivalent to $\xi(\chi) = \{2\}$. For functions $f = \sum_{i=0}^s \alpha_i \gamma_{2i}$ we can determine $\xi(f)$ and hence $\text{inv}(f)$ by using Theorem 2.3. For a polynomial $g \in \mathbb{F}_2[X]$ with $g(0) = 1$ we define the order $\text{ord}(g)$ as the smallest positive integer ℓ such that g divides $X^\ell + 1$. For more information on the order see [LN97].

Theorem 2.8. *Let $f = \sum_{i=0}^s \alpha_i \gamma_{2i}$ with $\alpha_0 = 1$ and $F(X) = \varphi(f)$. Then*

$$\xi(f) = \{2 \text{ord}(g_j) : j \in \{1, \dots, t\}\}$$

where $F(X) = g_1^{e_1} \cdots g_t^{e_t}$ is the factorization of F into irreducible factors.

3 Quasi-conserved functions and the polynomial composition property

We now set out to generalize the previous results. An important observation is that the key result of Section 2 is Lemma 2.1. Essentially any later results then follow from the correspondence between the functions and polynomials.

We start by observing that many shift-invariant functions can be described by a so-called (complementing) landscape, see for example [TM90]. Let us consider the function χ , take $x \in \mathbb{F}_2^n$ and set $y = \chi(x)$. It then follows that $y_i \in \{x_i, x_i + 1\}$ for each $i \in \{0, \dots, n-1\}$, hence to know y it is enough to know for which i we have $y_i = x_i + 1$. Note that from the definition of χ we have $y_i = x_i + (1 + x_{i+1})x_{i+2}$ and therefore $y_i = x_i + 1$ if and only if $(x_i, x_{i+1}, x_{i+2}) = (x_i, 0, 1)$. We then say that x matches the landscape *01 at position i and that χ is given by the landscape *01.

Note that in the following we work with vectors $x \in \mathbb{F}_2^{\mathbb{Z}}$, where $\mathbb{F}_2^{\mathbb{Z}}$ is the set of functions $\mathbb{Z} \rightarrow \mathbb{F}_2$, which we view as infinite vectors in both directions. This is mostly to ignore some technicalities with too small n , say $n < k + m$.

Here, shift-invariant means that $f(S(x)) = S(f(x))$ for all $x \in \mathbb{F}_2^{\mathbb{Z}}$ where $S : \mathbb{F}_2^{\mathbb{Z}} \rightarrow \mathbb{F}_2^{\mathbb{Z}}$ is the left-shift defined by $S : x \mapsto y$ with $y_i = x_{i+1}$ for all $i \in \mathbb{Z}$. The results that we show can still be applied to functions on $\mathbb{F}_2^n$.

Definition 3.1. A landscape c is a finite string consisting of the symbols $0, 1, _$, i.e. $c \in \{0, 1, _ \}^*$.

One symbol of c is highlighted. We index c in such a way that this occurs at position 0, i.e. $c = a_{-k}a_{-k+1} \dots a_{-1}\hat{a}_0a_1a_2 \dots a_m$ where $\hat{a}_0$ is the highlighted symbol and $a_\ell \in \{0, 1, _ \}$ for $\ell \in \{-k, \dots, m\}$. If $a_0 = _$, then instead of $\hat{_}$ we write $*$.

To a landscape $c = a_{-k}a_{-k+1} \dots a_{-1}\hat{a}_0a_1a_2 \dots a_m$ we associate a shift-invariant function $c : \mathbb{F}_2^{\mathbb{Z}} \rightarrow \mathbb{F}_2^{\mathbb{Z}}$ which is defined by $x \mapsto y$ with $y_i = 1$ if and only if

$$(x_{i-k}, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_{i+m}) = (a_{-k}, \dots, a_{-1}, \hat{a}_0, a_1, \dots, a_m) \quad (1)$$

where an equality of the form $x_j = _$ means that $x_j \in \mathbb{F}_2$ can be arbitrary. If (1) holds we say that x matches the landscape c at position i .

For a landscape $c = a_{-k}a_{-k+1} \dots a_{-1}\hat{a}_0a_1a_2 \dots a_m$ we define its support $\text{supp}(c)$ by

$$\text{supp}(c) = \{j \in \{-k, \dots, m\} : a_j \neq _ \}.$$

With this definition we have $\chi = \text{id} + c$ where $c = *01$. Note that this is somewhat different to the existing literature (e.g. [Dae95]) in which it is written that $\chi = *01$. Our notation is more convenient for us here.

Remark 3.2. We can recover the finite-dimensional setting $\mathbb{F}_2^n$ from $\mathbb{F}_2^{\mathbb{Z}}$ as follows:

- We can view c as a function $\mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ by computing all indices in (1) modulo n .
- More generally, given any shift-invariant function $f : \mathbb{F}_2^{\mathbb{Z}} \rightarrow \mathbb{F}_2^{\mathbb{Z}}$ with coordinate function $f(x)_i = g(x_{i-k}, \dots, x_{i+m})$ for some Boolean function $g : \mathbb{F}_2^{k+m+1} \rightarrow \mathbb{F}_2$ we can recover $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ by computing all indices appearing in g modulo n .
- Alternatively, we consider the subspace

$$\mathcal{P}_n = \{x \in \mathbb{F}_2^{\mathbb{Z}} : x_i = x_{i+n} \text{ for all } i \in \mathbb{Z}\} = \{x \in \mathbb{F}_2^{\mathbb{Z}} : x = S^n x\} \leq \mathbb{F}_2^{\mathbb{Z}}$$

of all vectors with period n , which is isomorphic to $\mathbb{F}_2^n$. For each shift-invariant function $f : \mathbb{F}_2^{\mathbb{Z}} \rightarrow \mathbb{F}_2^{\mathbb{Z}}$ we have the restriction $f : \mathcal{P}_n \rightarrow \mathcal{P}_n$. Identifying $\mathcal{P}_n$ with $\mathbb{F}_2^n$ we thus obtain shift-invariant functions $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$.

One can quickly check that these perspectives give equivalent results.

Remark 3.3. One can quickly translate between the landscape notation of a shift-invariant function and the notation using S and $\odot$ introduced at the beginning of Section 2. If $c = a_{-k} \dots a_m$, then

$$c = A_{-k} \odot \dots \odot A_m \quad (2)$$

where $A_j : \mathbb{F}_2^{\mathbb{Z}} \rightarrow \mathbb{F}_2^{\mathbb{Z}}$ are given by

$$A_j = \begin{cases} S^j & \text{if } a_j = 1, \\ \mathbb{1} + S^j & \text{if } a_j = 0, \\ \mathbb{1} & \text{if } a_j = -. \end{cases}$$

For example, $*01 = (\mathbb{1} + S) \odot S^2$ and $0*10 = (\mathbb{1} + S^{-1}) \odot S \odot (\mathbb{1} + S^2)$.

Consider the landscape $c = 0*10$. The associated function $f = \text{id} + c$ was found by Patt [AP72, Pat71]. It is a permutation on $\mathbb{F}_2^n$ for any n and on $\mathbb{F}_2^{\mathbb{Z}}$. The argument for this is simple. We observe that if $x \in \mathbb{F}_2^{\mathbb{Z}}$ matches c at position i , i.e. $(x_{i-1}, x_i, x_{i+1}, x_{i+2}) = (0, x_i, 1, 0)$, then x does not match c at any of the positions $i-1, i+1, i+2$. Therefore if we set $y = f(x)$, then $(y_{i-1}, y_i, y_{i+1}, y_{i+2}) = (0, x_i + 1, 1, 0)$ and then by the same argument $z = f(y) = x$. So in fact f is an involution on $\mathbb{F}_2^n$ and $\mathbb{F}_2^{\mathbb{Z}}$.

In contrast to this for the landscape $c = *01$ we have that if x matches c at position i , then it does not match c at position $i+1$ but may match c at position $i+2$. In other words, if $y = \chi(x)$ and $y_i = x_i + 1$, i.e. $(x_i, x_{i+1}, x_{i+2}) = (x_i, 0, 1)$ then $(y_i, y_{i+1}, y_{i+2}) = (x_i + 1, 0, y_{i+2})$. In particular, the entry at position $i+1$ did not change under application of χ . We generalize this to other landscapes for which it also holds that at most one of the relevant entries may be changed under application of the function.

Definition 3.4. Let c be a landscape. We say that c is a quasi-conserved landscape, if there exists one index $q \in \text{supp}(c)$ such that whenever $x \in \mathbb{F}_2^{\mathbb{Z}}$ matches c at position i , then x does not match c at all positions $i + \ell$ for $\ell \in \text{supp}(c) \setminus \{q\}$. Equivalently, setting $f = \text{id} + c$ and $y = f(x)$, whenever x matches c at position i , then $y_i = x_i + 1$ but $y_{i+\ell} = x_{i+\ell}$ for all $\ell \in \text{supp}(c) \setminus \{q\}$.

The previous discussion shows that $*01$ is a quasi-conserved landscape. In fact, a lot of the landscapes that are considered in the literature are captured by this definition.

Example 3.5. In [LYZ⁺26] the functions $\chi_{n,m} : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ were introduced, which are defined by

$$\chi_{n,m}(x)_i = x_i + (1 + x_{i+1})(1 + x_{i+2}) \cdots (1 + x_{i+m-1})x_{i+m}$$

for $i \in \{0, \dots, n-1\}$. Note that $\chi_{n,m} = \text{id} + c$ for the landscape $c = *0 \dots 01$. This landscape is quasi-conserved. The results of [LYZ⁺26] generalize those

of [KK24] to the case of the landscape $c = *0 \dots 01$. As our results generalize these same results to all quasi-conserved landscapes, our results also cover the results of [LYZ⁺26].

In Table A.1 of Daemen’s thesis [Dae95] there are 24 landscapes given. Of these, 15 are quasi-conserved, 7 are conserved and 2 are neither, see Table 1. In particular our results can explain many of the properties of these 15 quasi-conserved functions.

In [LDS⁺26] the problem of finding inverses of shift-invariant transformation in a systematic way is further studied. Among others they find the inverses of the functions given by the landscapes $*01$, $*011$ (which is used in Monolith [GKL⁺24]) and $0*01$. In [LDS⁺26, Table 1] they give more examples of functions for which the tools in their paper can compute the inverse function. In fact, all of these correspond to landscapes that are contained in [Dae95, Table A.1] and are either conserved or quasi-conserved. As our results also enable to find the inverse of any function given by a quasi-conserved landscape, our results imply theirs.

Landscape	Property	q if quasi-conserved
$*01$	quasi-conserved	2
$0*01$	quasi-conserved	2
$*001$	quasi-conserved	3
$*100$	quasi-conserved	3
$0*_10$	quasi-conserved	2
$*01_1$	neither	-
$*0_01$	quasi-conserved	4
$*01_0$	quasi-conserved	4
$0_ \dots _ *1_ \dots _ 0$	quasi-conserved	1
$*0100$	neither	-
$0*010$	quasi-conserved	2
$0*001$	quasi-conserved	3
$0*100$	quasi-conserved	3
$0*101$	quasi-conserved	3
$*0111$	quasi-conserved	4
$*0001$	quasi-conserved	4
$*0011$	quasi-conserved	4
$0*01_0$	quasi-conserved	4
$0*011$	conserved	-
$0_ *_11_0$	conserved	-
$0*1_ \dots _ 10$	conserved	-
$01*01$	conserved	-
$0_ \dots _ *1_ \dots _ 10$	conserved	-
$0_ *100$	conserved	-

Table 1: The landscapes from Table A.1 in [Dae95]

Given a landscape c we can define functions γ_j that generalize the functions γ_{2k} from [KK24, KK25] and $\theta_{m,k}$ from [LYZ⁺26].

Definition 3.6. Let $c = a_{-k} \dots a_m$ be a landscape and take $q \in \text{supp}(c)$ arbitrary. Define the landscape $c^{(q)}$ by omitting a_q from c , i.e.

$$c^{(q)} = a_{-k} \dots a_{q-1} a_{q+1} \dots a_m.$$

We define functions $\gamma_j^{(c,q)} : \mathbb{F}_2^\mathbb{Z} \rightarrow \mathbb{F}_2^\mathbb{Z}$ for $j \in \mathbb{N}_0$ by $\gamma_0^{(c,q)} = \text{id}$, $\gamma_1^{(c,q)} = c$ and

$$\gamma_{j+1}^{(c,q)} = c^{(q)} \odot S^q \gamma_j^{(c,q)}$$

for $j \geq 1$. If c and q are clear from context then we simply write γ_j . We set

$$\Gamma^{(c,q)} = \text{span}\{\gamma_0, \gamma_1, \gamma_2, \dots\}.$$

Example 3.7. We consider two examples.

Let first $c = *01 = (\mathbb{1} + S) \odot S^2$. Then $q = 2$ so that $c^{(q)} = *0_- = \mathbb{1} + S$ and therefore

$$\begin{aligned} \gamma_2 &= c^{(q)} \odot S^2(c) = (\mathbb{1} + S) \odot S^2[(\mathbb{1} + S) \odot S^2] \\ &= (\mathbb{1} + S) \odot (\mathbb{1} + S^3) \odot S^4, \\ \gamma_3 &= c^{(q)} \odot S^2(\gamma_2) = (\mathbb{1} + S) \odot S^2[(\mathbb{1} + S) \odot (\mathbb{1} + S^3) \odot S^4] \\ &= (\mathbb{1} + S) \odot (\mathbb{1} + S^3) \odot (\mathbb{1} + S^5) \odot S^6. \end{aligned}$$

Observe that these are the functions γ_{2k} from Section 2. We can also write this computation in landscape notation:

$$\begin{aligned} \gamma_2 &= c^{(q)} \odot S^2(c) = *0_- \odot S^2(*01) = *0 \odot *_-_01 = *0_-01, \\ \gamma_3 &= c^{(q)} \odot S^2(\gamma_2) = *0_- \odot *_-_0_01 = *0_-0_01. \end{aligned}$$

Let now $c = *0111 = (\mathbb{1} + S) \odot S^2 \odot S^3 \odot S^4$. Then $q = 4$ and $c^{(q)} = *011_- = (\mathbb{1} + S) \odot S^2 \odot S^3$ and therefore

$$\begin{aligned} \gamma_2 &= c^{(q)} \odot S^4(c) = (\mathbb{1} + S) \odot S^2 \odot S^3 \odot S^4 \odot S^4[(\mathbb{1} + S) \odot S^2 \odot S^3] \\ &= (\mathbb{1} + S) \odot S^2 \odot S^3 \odot (\mathbb{1} + S^5) \odot S^6 \odot S^7 \odot S^8, \\ \gamma_3 &= (\mathbb{1} + S) \odot S^2 \odot S^3 \odot (\mathbb{1} + S^5) \odot S^6 \odot S^7 \odot (\mathbb{1} + S^9) \odot S^{10} \odot S^{11} \odot S^{12}. \end{aligned}$$

In landscape notation:

$$\begin{aligned} \gamma_2 &= *011_-0111 \\ \gamma_3 &= *011_-011_0111. \end{aligned}$$

We can then define the property that the functions γ_j satisfy the result of Lemma 2.1:

Definition 3.8. Let c be a landscape and $q \in \text{supp}(c)$. We say that (c, q) satisfies the polynomial composition property if for every $j \geq 0$ and $\alpha_0, \dots, \alpha_s \in \mathbb{F}_2$ with $\alpha_0 = 1$ it holds that

$$\gamma_j^{(c,q)} \circ \left(\gamma_0^{(c,q)} + \sum_{i=1}^s \alpha_i \gamma_i^{(c,q)} \right) = \sum_{i=0}^s \alpha_i \gamma_{i+j}^{(c,q)}.$$

The theorem we prove is the following:

Theorem 3.9. *Let c be a landscape. Then the following are equivalent:*

1. *c is quasi-conserved with special index q .*
2. *(c, q) satisfies the polynomial composition property.*

We cannot prove Theorem 3.9 here, hence we only give some intuition. One can show that c is quasi-conserved if and only if $S^\ell c \odot c$ vanishes for all $\ell \in \text{supp}(c) \setminus \{q\}$. The expression $\gamma_j \circ (\sum_{i=0}^s \alpha_i \gamma_i)$ is equivalent to a product whose expansion contains terms of the form $S^\ell \gamma_k \odot \gamma_t$ for some parameters ℓ, k, t . By using the previous property one can show that enough of these terms vanish so that the end result is $\sum_{i=0}^s \alpha_i \gamma_{i+j}$. The other direction of the theorem follows by similar considerations.

Theorem 3.9 implies that the functions γ_j for any quasi-conserved landscape satisfy Lemma 2.1 which all results in [KK24, KK25] are based on. Hence, we can prove analogous results of all theorems in Section 2 for all quasi-conserved landscapes, for instance those landscapes mentioned in Example 3.5. In particular, given a fixed quasi-conserved landscape c and a function $f \in \gamma_0 + \text{span}\{\gamma_1, \gamma_2, \dots\}$ we can then answer the following questions, which are crucial for understanding and analyzing cryptographic primitives based on shift-invariant functions:

1. For which n is f a permutation on $\mathbb{F}_2^n$?
2. If f is a permutation on $\mathbb{F}_2^n$, then what is a formula for the inverse of f ? What is the degree of the inverse?
3. If f is a permutation on $\mathbb{F}_2^n$, then what is the order of f ?
4. What is a formula for f^k ?
5. Information on the cycle structure of f ?

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